

AgriEdge Or-Mange Ltd: Salesforce CRM Order Management System

Project Documentation

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Project Overview

AgriEdge Or-Mange Ltd, a key player in the agriculture and food production sector, aims to modernize its order management system (OMS) by leveraging Salesforce CRM. The system is designed to streamline order processing, provide real-time inventory tracking, and enhance customer service. The CRM will automate various processes, reduce manual errors, and integrate smoothly with existing service channels to ensure customer satisfaction and operational efficiency.

Objectives

- Automate and streamline order processing workflows.
 - Enable real-time inventory visibility to avoid stockouts and overstock.
 - Provide seamless integration with customer service platforms (email, phone, portal).
 - Improve data accuracy and operational reporting using analytics.
 - Ensure data security and compliance with regulatory standards.
 - Enhance scalability for future enhancements like chatbot support or AI insights.
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Project Phases

Phase 1: Requirement Analysis & Planning

Understanding Business Requirements

- Manual order entries leading to human error.
- Lack of real-time stock data resulting in poor inventory decisions.
- Disconnected support channels making customer service inefficient.

Defining Project Scope and Objectives

- Replace manual processes with automated workflows.
- Centralize customer, order, and inventory data within Salesforce.
- Connect customer service channels into the CRM.
- Track field history, automate status changes, and ensure order traceability.

Design Data Model and Security Model

- Custom Objects:
 - AgriEdge_Order__c – Tracks order details (status, items, delivery date).
 - AgriEdge_Shipment__c – For shipment tracking.

- Product__c, Inventory__c, Customer__c
- Relationships: Lookup and Master-detail relationships as needed.
- Security: Profiles and Role Hierarchy configured with permission sets to manage access.

Phase 2: Salesforce Development – Backend & Configurations

Setup Environment & DevOps Workflow

- Used Developer Sandbox for initial development.
- Version control using change sets and manual deployment (Change Set-based strategy).

Customizations:

- Objects & Fields for Orders, Shipments, Products, Inventory.
- Validation Rules:
 - Prevent order submission if stock is insufficient.
 - Ensure delivery date is later than order date.
- Automation:
 - **Flows:** Auto-create Tasks for new orders.
 - **Process Builder:** Auto-update order status when shipment is created.
 - **Approval Process:** Manager approval for large volume orders.

Apex Classes and Triggers:

- OrderTaskCreator: Auto-create tasks when an order is inserted.
- ShipmentStatusUpdater: Updates order status to "Shipped" upon shipment record creation.
- **Test Classes:** Created with >75% coverage for all Apex logic.

Phase 3: UI/UX Development & Customization

Lightning App Setup

- Custom App: "AgriEdge OMS" via App Manager.

Page Layouts & Dynamic Forms

- Customized layouts for Orders and Products based on profiles.

- Dynamic fields shown depending on order status.

User Management

- Roles: Admin, Inventory Manager, Customer Support.
- Profiles: Field-level security set per profile.
- Permission Sets: For report/dashboard access.

Reports & Dashboards

- Dashboard for inventory levels, pending orders, and shipment statuses.
- Reports on top customers, average delivery times.

Lightning Pages & (Optional) LWC

- Lightning pages created per object.

Phase 4: Data Migration, Testing & Security

Data Loading

- Used Data Loader for bulk upload of Products and Inventory.
- Used Data Import Wizard for Accounts and Contacts.

Security Settings

- Enabled Field History Tracking on AgriEdge_Order__c.
- Configured Duplicate Rules for Accounts and Contacts.
- Matching Rules for customer identification.

Profiles & Sharing

- Configured Role Hierarchy: Admin > Manager > Agent.
- Used Organization-Wide Defaults and Sharing Rules.

Test Classes

- All Apex Classes tested with positive & negative test cases.

Phase 5: Deployment, Documentation & Maintenance

Deployment Strategy

- Used **Change Sets** for moving components from Sandbox to Production.
- Deployment validated via test run.

Maintenance Plan

- Monthly review of inventory reports.
- Support case logs monitored via Service Console.
- Troubleshooting via debug logs and error e-mail alerts.

Troubleshooting Guide

- Common Errors:
 - "Inventory Insufficient" → Check Product Stock field.
 - Approval stuck → Reassign via Approval History.
 - Logs to monitor: OrderTaskCreator, ShipmentStatusUpdater.
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Conclusion

The Salesforce-powered Order Management System for AgriEdge Or-Mange Ltd successfully transforms a previously manual and error-prone process into an automated, efficient, and responsive system. By centralizing data, automating workflows, and enhancing visibility, the system meets business needs while setting a strong foundation for future enhancements such as chatbot integration, AI-powered order predictions, and IoT-based stock updates.